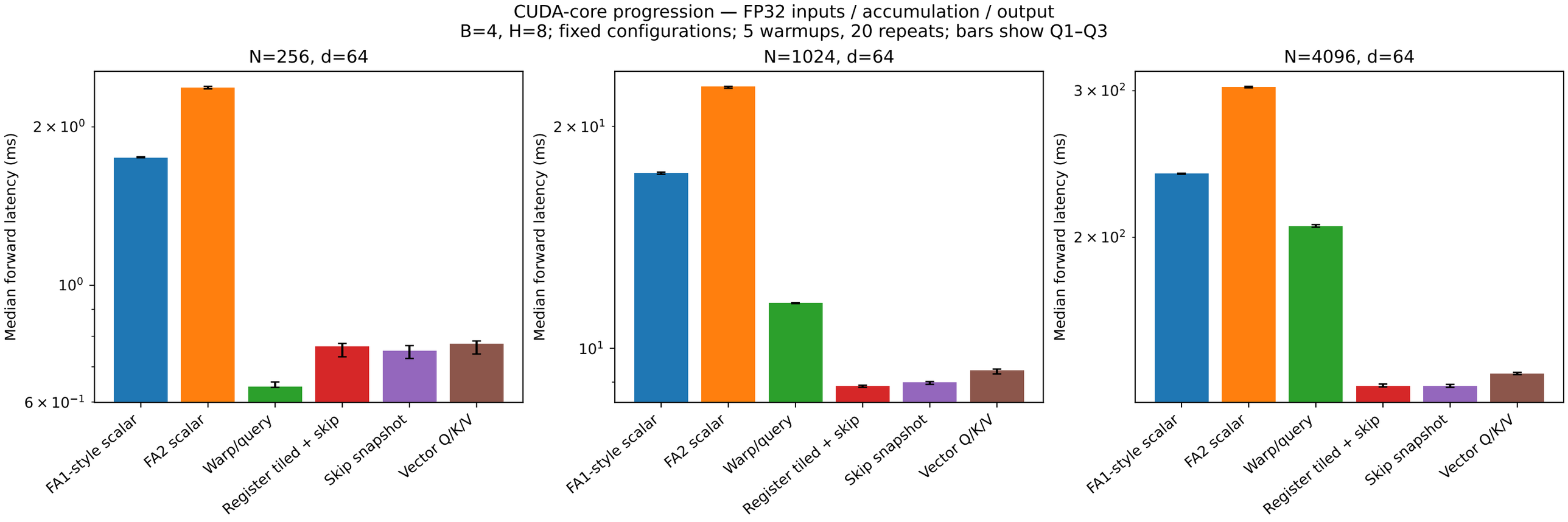
CUDA-core progression — FP32 inputs / accumulation / output
B=4, H=8; fixed configurations; 5 warmups, 20 repeats; bars show Q1–Q3
N=256, d=64
N=1024, d=64
N=4096, d=64
Median forward latency (ms)
2×10^{0}
10^{0}
6×10^{-1}
2×10^{1}
10^{1}
3×10^{2}
2×10^{2}
FA1-style scalar
FA2 scalar
Warp/query
Register tiled + skip
Skip snapshot
Vector Q/K/V